

Difference-in-Differences

SW Chapter 10 (Part 2)

ECON3500: Econometrics and Applications
Spring 2026

Where We Are

Last time, we learned how **panel data** lets us control for unobserved, time-invariant factors:

- **First differencing**: Subtract one period from another — α_i drops out
- **Fixed effects**: Demean within each entity — same result, works with $T > 2$
- Both eliminate OVB from **time-invariant** unobservables (Ch 9, Threat #1)

Today: a specific research design that uses these tools to **credibly estimate** the effects of policies and events — under clearly stated assumptions.



The Big Idea

Difference-in-differences (DiD) combines a **treatment/control comparison** with a **before/after comparison** to address confounding — even without randomization — *if* the **parallel trends assumption** holds.

The Problem with Simple Comparisons

Motivating Example: Garbage Incinerator

Question: What is the effect of a garbage incinerator on nearby housing prices?

After the incinerator was built:



$$\widehat{r_{price}} = 101,308 - 30,688 \cdot nearinc$$

Houses near the incinerator sell for ~\$30k less. Did the incinerator cause this?

Not So Fast...

Look at the relationship **before** the incinerator was built:



$$\widehat{rprice} = 82,517 - 18,824 \cdot nearinc$$

The incinerator was built in a place where housing prices were **already depressed!**

The \$30k gap reflects both the incinerator effect **and** pre-existing differences.

Two Flawed Comparisons

Cross-Sectional Comparison (After Only)

Compare near vs. far houses **after** the incinerator:
−\$30,688

Problem: Near-incinerator houses were **already cheaper**. Location characteristics (neighborhood quality, amenities) are **confounders** — they affect both proximity to the incinerator *and* housing prices. In DAG terms (Ch 8b): there's an open **backdoor path** through location characteristics.

Before/After Comparison (Treatment Only)

Compare near-incinerator houses before vs. after:
+\$18,790

Problem: Housing prices were **rising everywhere**. Time trends (inflation, economic growth) affect prices regardless of the incinerator. We're mixing the treatment effect with a common time shock.

Both comparisons violate **conditional mean independence** ($E(u_i | X_i) \neq 0$, Ch 9). We need a method

that handles **both** problems simultaneously.

Difference-in-Differences

The Core Logic



Step 1: How much did the gap change over time?

$$\hat{\delta}_1 = \underbrace{(-30,688)}_{\text{after gap}} - \underbrace{(-18,824)}_{\text{before gap}} = -11,864$$

The DiD estimate suggests the incinerator reduced nearby prices by ~\$12k — not \$30k. (We'll discuss the assumption needed for a **causal** interpretation shortly.)

The 2x2 Table

The name says it all — it's the **difference** of two **differences**:

	Before	After	(After - Before)
Control (far from incinerator)	\$82,517	\$101,308	+\$18,790
Treatment (near incinerator)	\$63,693	\$70,619	+\$6,927
Δ (Treatment - Control)	-\$18,824	-\$30,688	-\$11,864

Reading the bottom-right cell:

$$\underbrace{(70,619 - 63,693)}_{\Delta \text{ treatment}} - \underbrace{(101,308 - 82,517)}_{\Delta \text{ control}}$$

DiD Graphically

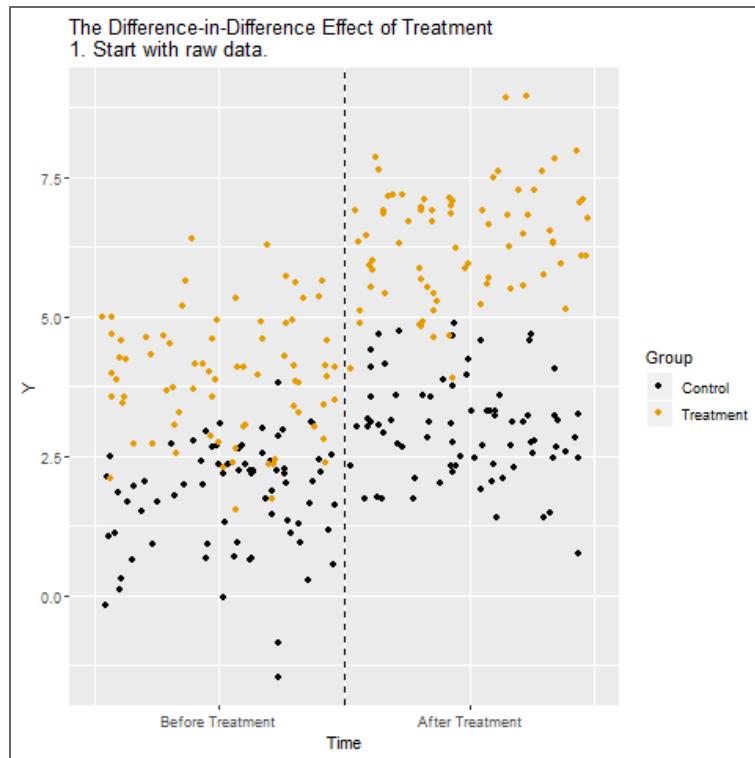


The dashed line shows where near-incinerator prices *would have been* if they followed the same trend as far-away prices.

The **DiD estimate** is the vertical gap between:

- What **actually happened** to the treatment group
- The **counterfactual**: what would have happened without treatment

DiD Animated



Animation by Nick Huntington-Klein ([The Effect](#))

Watch the steps:

1. Raw data: two groups over time
2. Collapse to group means
3. Measure the control group's time trend
4. **Subtract** that trend from the treatment group
5. What remains = the **DiD estimate**

The key: use the control group's trajectory to build the **counterfactual** for the treatment group.

The Counterfactual

! What DiD Really Estimates

DiD doesn't compare treatment and control groups directly. It asks:

*How much did the treatment group change **relative to what would have happened without treatment?***

The control group's trajectory provides the counterfactual — but only if parallel trends holds.

This is why DiD is so useful for **policy evaluation**:

- Treatment and control groups can have **very different levels** (the backdoor path from time-invariant confounders is closed by differencing — Ch 8b)
- They just need to be on **similar trajectories** before the policy hits
- The identifying assumption is about **trends**, not **levels**

The Regression Framework

DiD as a Regression

We can capture the entire DiD logic in one regression:

$$rprice_i = \beta_0 + \delta_0 \cdot after_i + \beta_1 \cdot nearinc$$

	Before ()	After ()	After Before
Control (far)	β_0	$\beta_0 + \delta_0$	δ_0
Treatment (near)	$\beta_0 + \beta_1$	$\beta_0 + \delta_0 + \beta_1 + \delta_1$	$\delta_0 + \delta_1$
Treat – Control	β_1	$\beta_1 + \delta_1$	δ_1

δ_1 — the coefficient on the **interaction term** — is the DiD estimate.

Interpreting Each Piece

$$y_i = \underbrace{\beta_0}_{\text{baseline}} + \underbrace{\delta_0}_{\text{time trend}} \cdot \underbrace{after_i}_{\text{time trend}} + \underbrace{\beta_1}_{\text{group difference}} \cdot \underbrace{treated_i}_{\text{group difference}}$$

Coefficient	Meaning
β_0	Average outcome for control group, before
δ_0	How the control group changed over time (time trend)
β_1	Pre-existing difference between groups
δ_1	The DiD estimate — causal <i>if</i> parallel trends holds



Knowledge Check

In the regression

$rprice_i = 82,517 + 18,790 \cdot after_i - 18,824 \cdot neari$
 , what does 18,790 represent?

...

The increase in housing prices among far-away houses over time — the common time trend.

Adding Control Variables

DiD works fine with additional covariates:

$$y_{it} = \beta_0 + \delta_0 \cdot after_t + \beta_1 \cdot treated_i + \delta_1$$

Why add controls? Two distinct reasons:

1. **Precision:** Controls **reduce residual variance** → tighter standard errors, even if parallel trends already holds unconditionally.
2. **Credibility:** Parallel trends may only hold **conditional on covariates**. If treatment and control groups differ on observables that predict trends, controlling for those variables can make parallel trends more plausible.



Controls Can Matter for Identification

Unlike in an RCT, where controls are purely about precision, in DiD the parallel trends assumption may only hold *after conditioning* on observables. If so, omitting those variables is **OVB** (Ch 9, Threat #1) — the controls are closing a **backdoor path** (Ch 8b).

Example: Controlling for house characteristics (square footage, bedrooms, lot size) when estimating the incinerator effect. If newer/larger homes were built farther

from the site during this period, the controls aren't just adding precision — they're needed for parallel trends.

The Parallel Trends Assumption

The Key Assumption

! Parallel Trends Assumption (*common trends assumption in SW*)

In the **absence of treatment**, the difference between treatment and control groups would have remained **constant over time**.

Equivalently: both groups would have followed **parallel trajectories**.

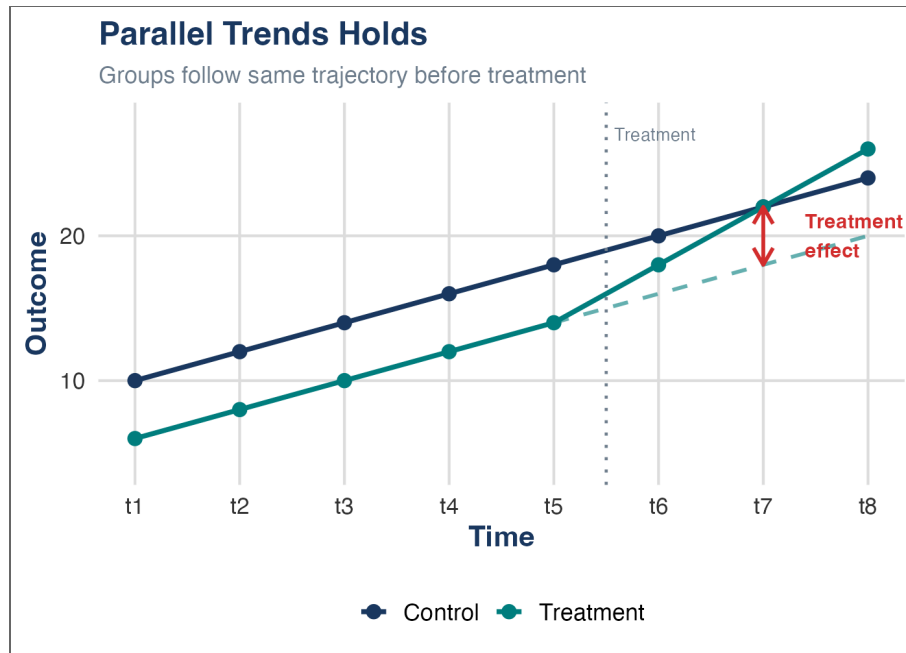
What this **allows**:

- Treatment and control groups can have **different levels** of the outcome
- There can be **time-invariant** unobserved confounders — DiD differences them out (just like FE)

What this **requires**:

- No **time-varying confounders** that differentially affect the two groups
- In Ch 9 terms: after the DiD structure removes time-invariant OVB, no remaining threat to $E(u_{it} | X_{it}) = 0$

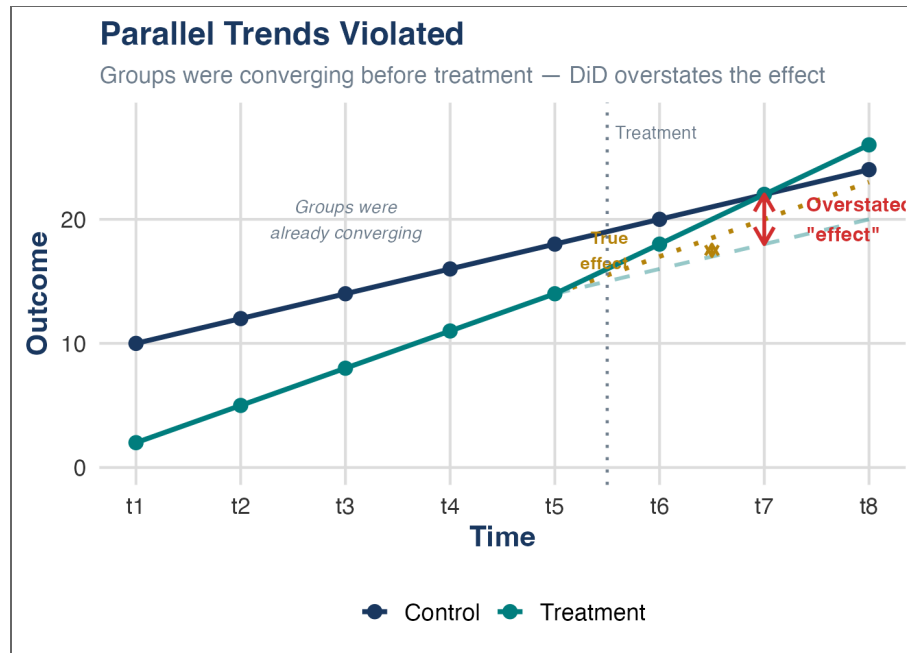
Parallel Trends Holds



Groups follow the same trajectory before treatment. The treatment effect is the gap between the actual outcome and the counterfactual (dashed line).

The dashed line = the **counterfactual**: where the treatment group would have been if it followed the control group's trend.

Parallel Trends Violated



Groups were converging before treatment. DiD attributes the continued convergence to the treatment — overstating the true effect.

The gold dotted line = the **true counterfactual** (continuing convergence). The dashed line = what DiD **incorrectly assumes**. The difference between the two counterfactuals is the bias.

What Would Violate Parallel Trends?

In the Incinerator Example

- A new highway built near the incinerator site at the same time → affects near-incinerator prices but not far-away prices
- A neighborhood revitalization program targeting the area → changes near-incinerator prices independent of the incinerator
- Differential migration patterns (people leaving the area because the incinerator was *announced*)

More Generally — Time-Varying OVB (Ch 9, Threat #1)

Any factor that:

1. **Changes over time** (not absorbed by entity or time FE)
2. Affects treatment and control groups **differently**
3. Coincides with the timing of treatment

In DAG terms (Ch 8b): these are **time-varying confounders** — variables that open a backdoor path between treatment status and the outcome *that*

DiD cannot close. Entity FE closes backdoor paths from time-invariant confounders; parallel trends assumes no time-varying backdoor paths remain.

Assessing Parallel Trends

We can never **prove** parallel trends — it's about what **would have happened** without treatment, which we don't observe.

But we can look for **supporting evidence**:

1. Pre-treatment trend test: Plot the outcome for both groups over time. Do they move in parallel *before* treatment?

2. Placebo/falsification tests:

- Run DiD using a **fake treatment date** (before the real one). If you find an “effect,” something else is going on.
- Run DiD on an outcome that **should not be affected** by treatment. A significant result suggests a violation.

3. Event study plot: Estimate treatment effects for each period relative to treatment. Pre-treatment coefficients should be **near zero and flat**.

The Event Study Plot

An **event study** estimates separate effects for each period relative to treatment:

$$y_{it} = \alpha_i + \lambda_t + \sum_{k \neq -1} \delta_k \cdot D_{it}^k + u_{it}$$

where $D_{it}^k = 1$ if unit i is k periods from treatment at time t .

What to look for:

- **Pre-treatment coefficients** ($k < 0$): Should be close to zero and statistically insignificant → supports parallel trends
- **Post-treatment coefficients** ($k \geq 0$): Show the dynamic treatment effect over time
- The reference period ($k = -1$) is normalized to zero



Why Event Studies Matter

Event study plots have become nearly **mandatory** in applied DiD papers. Referees expect to see them. They are the most credible way to support the parallel trends assumption.

Real-World DiD Examples

Classic Example: Card & Krueger (1994)

Question: Does raising the minimum wage reduce employment?

Setting:

- New Jersey raised its minimum wage from \$4.25 to \$5.05 in April 1992
- Pennsylvania did not change its minimum wage
- Surveyed fast-food restaurants in both states, before and after

	Before	After	
NJ (treatment)	20.44 FTE	21.03 FTE	+0.59
PA (control)	23.33 FTE	21.17 FTE	-\$2.16
NJ – PA			+2.76

Result: The DiD estimate suggests employment **increased** in NJ relative to PA — the opposite of the standard prediction. The causal interpretation rests on parallel trends: that NJ and PA employment would have moved together absent the wage increase. One of the most influential papers in labor economics.

Why Is Card & Krueger Compelling?

Good DiD Design Features

1. **Sharp treatment:** NJ raised the minimum wage on a specific date; PA didn't
2. **Geographic neighbors:** NJ and PA share economic conditions → plausible parallel trends
3. **Same industry:** Fast food in both states faces similar demand shocks
4. **Pre-trends:** Employment was trending similarly before the increase

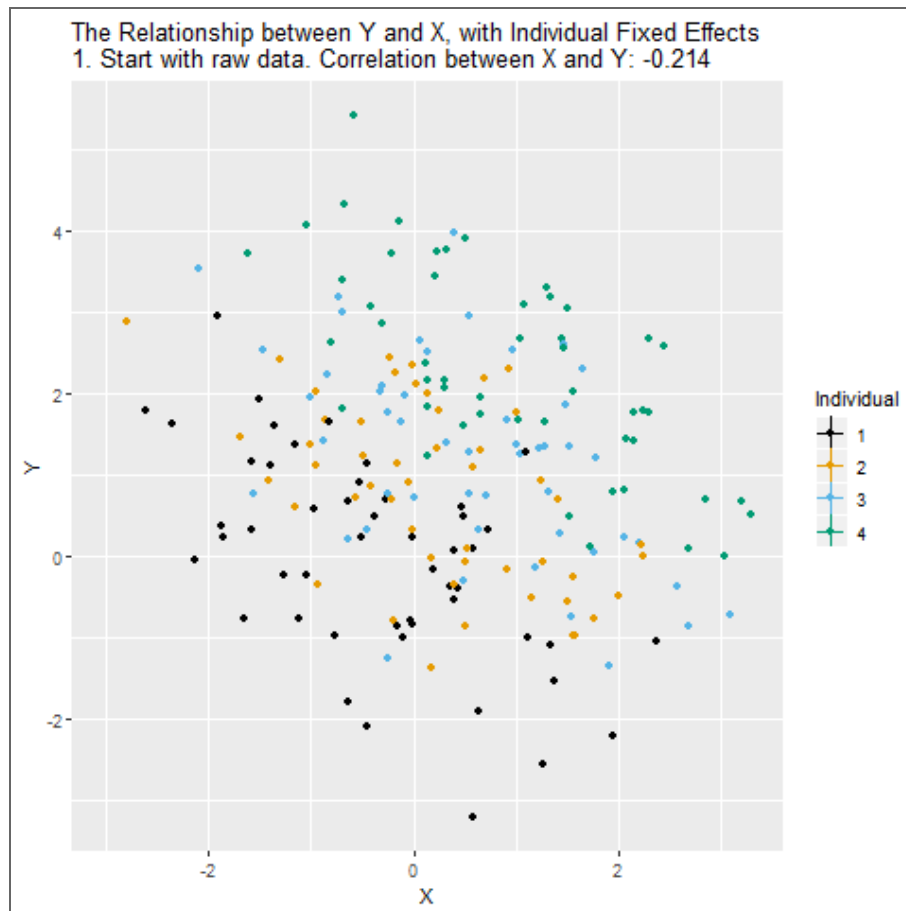
Potential Concerns (Ch 9 Threats in Disguise)

- Were NJ restaurants **anticipating** the wage increase? (announcement effects violate the sharp timing)
- Did PA have its own employment shocks unrelated to the minimum wage? (**time-varying confounders** — OVB)
- Are the two states similar *enough* for parallel trends?

- **Measurement error** (Ch 9, Threat #3):
Employment data came from phone surveys of managers — how accurate are the FTE counts?

Quick Recall: Fixed Effects Intuition

Remember what we learned last time about how fixed effects work:



Animation by Nick Huntington-Klein (**The Effect**)

DiD and fixed effects are closely related — DiD **uses** entity and time fixed effects, plus adds a specific treatment-timing structure (the interaction term). FE removes time-invariant confounders; the DiD design layer addresses the causal question *if* parallel trends holds.

Staggered Adoption and Recent Developments

Beyond Two Groups and Two Periods

The classic DiD has two groups and two periods. In practice, policies are often adopted by **different units at different times** — **staggered adoption**.

Example: States adopt a job training program in different years.

The natural approach: two-way fixed effects (TWFE).

$$y_{it} = \beta_1 \cdot treated_{it} + \alpha_i + \lambda_t + u_{it}$$

Recent research (roughly 2018–2024) has shown that this “obvious” approach can produce **severely misleading estimates** — including **wrong signs**.

Let’s see exactly how this happens.

A Concrete Example: Setup

Suppose a job training program has a **true effect** that **grows over time**:

- Year 1 after adoption: productivity rises by **+10**
- Year 2 after adoption: productivity rises by **+50** (the program grows dramatically)

Two states, three years. Without treatment, both would follow the same trend:

	2018	2019	2020
No-treatment baseline (both states)	100	105	110

- **State E (Early)**: adopts the program in **2019**
- **State L (Late)**: adopts the program in **2020**
- No pure control group exists (both eventually get treated)

What Actually Happens

	2018	2019	2020
State E (treats in 2019)	100	115	160
State L (treats in 2020)	100	105	120

- State E in 2019: $105 + 10 = 115$ (year 1 effect)
- State E in 2020: $110 + 50 = 160$ (year 2 effect — program grows dramatically)
- State L in 2020: $110 + 10 = 120$ (year 1 effect)

The true effect for State L in 2020 is **+10**. Can TWFE recover this?

What TWFE Does

With no pure control group, TWFE takes a **weighted average** of all possible 2×2 DiD comparisons — including contaminated ones.

Comparison 1 — valid: E treated vs. L as untreated control (2018→2019)

$$\underbrace{(115 - 100)}_{\Delta \text{State E}} - \underbrace{(105 - 100)}_{\Delta \text{State L}} = 15 - 5 = [+10]$$

$\Delta \text{State E}$

$\Delta \text{State L}$

Comparison 2 — contaminated: L treated vs. E **already-treated** as “control” (2019→2020)

$$\underbrace{(120 - 105)}_{\Delta \text{State L}} - \underbrace{(160 - 115)}_{\Delta \text{State E}} = 15 - 45 = [-30]$$

$\Delta \text{State L}$

$\Delta \text{State E}$

With equal-sized groups, TWFE weights these comparisons equally:

$$\hat{\beta}_{TWFE} = \frac{1}{2}(+10) + \frac{1}{2}(-30) = [-10].$$

The Sign Flipped!

! TWFE Says the Effect is -10

All true effects are **positive** (+10 or +50), but TWFE estimates **-10**. The sign flipped!

Why? State E's outcome rose by 45 from 2019→2020 — not from a trend, but because its **treatment effect grew** from +10 to +50. TWFE treats this jump as a “normal trend” in the control, making State L's +15 gain look like a relative **decline**.

i The Core Problem (Goodman-Bacon, 2021)

TWFE is a **weighted average** of all possible 2×2 DiD comparisons. When treatment effects **change over time**, comparisons using **already-treated units as controls** introduce bias — and can even get negative weights.

Solutions: New Estimators

Several estimators have been developed to handle staggered adoption correctly:

- **Callaway and Sant'Anna (2021):** Estimate group-time-specific effects, then aggregate. Only uses not-yet-treated or never-treated units as controls.
- **Sun and Abraham (2021):** Interaction-weighted estimator that corrects for heterogeneous effects across adoption cohorts.
- **de Chaisemartin and D'Haultfoeuille (2020):** Identifies which comparisons get negative weights and proposes robust alternatives.

All share one insight: **never use already-treated units as controls.**

What This Means for Practice



Practical Guidance

- **Classic 2×2 DiD** (two groups, two periods): TWFE is **fine**. The new concerns don't apply.
- **Staggered adoption** with **constant treatment effects**: TWFE is usually okay.
- **Staggered adoption** with **dynamic or heterogeneous effects**: Use a robust estimator.

How to check: Run both TWFE and a robust estimator. If results are similar, the standard approach is probably fine. If they diverge, trust the robust estimator.



Beyond ECON3500

You won't need to implement these in this course. But if you do applied research with staggered DiD (very common in policy evaluation), you should be aware of these issues. Stata packages: `csdid`, `did_multiplegt`, `eventstudyinteract`.

DiD in Stata

Estimating DiD

Option 1: Interaction regression

```
* Generate interaction term
gen after_treat = after * treated

* Estimate DiD
reg y after treated after_treat, robust
```

Option 2: Factor variable notation (preferred)

```
* Stata creates the interaction automatically
reg y i.after##i.treated, robust
```

The coefficient on `1.after#1.treated` is the DiD estimate $\hat{\delta}_1$.

DiD with Panel Data in Stata

With true panel data (same entities over time), combine DiD with fixed effects:

```
* Set panel structure
xtset entity_id year

* Entity + time FE with DiD
xtreg y treated_post i.year, fe vce(cluster entity_id)
```

Or equivalently, using `reghdfe`:

```
reghdfe y treated_post, absorb(entity_id year) vce(cluster entity_id)
```

Getting `reghdfe`

`reghdfe` (Correia, 2019) is a community-contributed command — install it with `ssc install reghdfe`. Stata 18+ added a built-in `absorb()` option to `regress`, but `reghdfe` remains the standard in applied work because it handles multi-way FE efficiently, reports useful diagnostics, and is what you'll see in published papers.

 **Don't Forget to Cluster!**

With panel data, **always** cluster standard errors at the entity level. Failing to do so overstates precision because of **serial correlation**.

Wrapping Up

DiD and the Internal Validity Framework

DiD is a direct response to the threats we catalogued in Ch 9:

Ch 9 Threat	How DiD Helps	What DiD Cannot Fix
OVB (Threat #1)	Eliminates time-invariant confounders (via FE/differencing)	Time-varying confounders that differentially affect groups
Wrong functional form (#2)	Flexible — can add covariates, nonlinearities	Misspecified treatment timing or group definitions
Measurement error (#3)	Not addressed	Attenuation bias in treatment or outcome still applies
Sample selection (#4)	Not addressed	Differential attrition (e.g., people move away <i>because</i>

Ch 9 Threat	How DiD Helps	What DiD Cannot Fix
		of treatment) creates collider bias (Ch 8b)
Simultaneous causality (#5)	Treatment timing helps — policy happens at a known date	If treatment is <i>responsive</i> to anticipated outcomes, timing is endogenous

In DAG terms: DiD closes backdoor paths from time-invariant confounders. Parallel trends assumes no **time-varying backdoor paths** remain open.

DiD Checklist

Before you trust a DiD estimate, ask:

- ☐ Is there a clear **treatment group** and **control group**?
- ☐ Is there a clear **before** and **after** period?
- ☐ Is **parallel trends** plausible? What evidence supports it?
- ☐ Are there **pre-trend tests** or an **event study plot**?
- ☐ Could any **time-varying confounder** have changed differentially at the same time? (Draw the DAG — Ch 8b)
- ☐ If staggered adoption: is TWFE appropriate, or do you need a robust estimator?
- ☐ Are standard errors **clustered** appropriately?

Key Takeaways

1. DiD combines a **cross-sectional** comparison with a **time** comparison — the causal interpretation requires the **parallel trends assumption**
2. The **interaction term** in the regression captures the DiD estimate ($\hat{\delta}_1$)
3. The **parallel trends assumption** is **untestable** (it's about a counterfactual), but you can look for supporting evidence via pre-trends and event studies
4. DiD addresses **time-invariant OVB** (Ch 9, Threat #1) by differencing it out — but **time-varying confounders** that differentially affect groups still threaten validity (open backdoor paths that DiD cannot close — Ch 8b)
5. With **staggered adoption**, standard TWFE can give misleading results — newer estimators fix this
6. Always **cluster standard errors** at the entity level with panel data



ECON3500 | Chapter 10b: Difference-in-Differences

Speaker notes